

P1293R1
Revision of P1293R0
2018-11-25
Mike Spertus, Symantec
mike_spertus@symantec.com
Nathan Wilson
nwilson20@gmail.com
Target: Library Evolution, Library

ostream_joiner

Summary

This paper updates the ostream_joiner proposal to reflect experience in Library Fundamentals V2 and changes in the core language since then. Specifically, we add a deduction guide to ensure that ostream_joiner is constructed correctly from a C string literal and remove the now-redundant make_ostream_joiner.

Deduction guides

With the addition of class template argument deduction in C++17, it seems desirable to get rid of the clumsy make_ostream_joiner as we did with class templates like boyer_moore_searcher. However, as Nathan Myers has pointed out, without any deduction guide, the decision to take the delimiter by reference means that

```
ostream_joiner j(cout, ", ");
```

produces a compile time error as the delimiter type is deduced as char const[3]. To fix this, add the following to the end of the class definition for ostream_joiner

Wording

Wording is relative to Library Fundamentals v2. In iterator.synopsis, remove the following

```
template <class DelimT, class charT = char, class traits = char_traits<charT> >  
    class ostream_joiner;  
template <class charT, class traits, class DelimT>  
    ostream_joiner<decay_t<&DelimT>, charT, traits>  
    make_ostream_joiner(basic_ostream<charT, traits>& os, DelimT&& delimiter);
```

At the end of the definition of class ostream_joiner in iterator.ostream.joiner, add the following.

```
};  
  
template<class DelimT, class CharT, class Traits>  
ostream_joiner(basic_ostream<CharT, Traits>, DelimT) -> ostream_joiner<DelimT, CharT, Traits>;
```

Remove the iterator.ostream.joiner.creation section

```
ostream_joiner creation function [iterator.ostream.joiner.creation]  
  
template <class charT, class traits, class DelimT>  
ostream_joiner<decay_t<DelimT>, charT, traits>  
make_ostream_joiner(basic_ostream<charT, traits>& os, DelimT&& delimiter);
```

Returns:

```
ostream_joiner<decay_t<DelimT>, charT, traits>(os, forward<DelimT>(delimiter));
```